

Functional Requirement Document

Here is the detailed Functional Requirement Document (FRD) based on the provided Business Requirement Document (BRD):

FR-DLT-001: Load Customer and Order Data to Delta Tables

Title: Load Customer and Order Data to Delta Tables

Description: The system must read source CSV data from volume and load it to Delta tables for customer and order data.

Preconditions:

CSV files are available at the specified locations:

`/Volumes/gen\_ai\_poc\_databrickscoe/sdlc\_wizard/customerdata` and
`/Volumes/gen\_ai\_poc\_databrickscoe/sdlc\_wizard/orderdata`

- The Delta tables for customer and order data do not exist or are empty

Main Flow / Functional Steps:

1. Read customer CSV data from `/Volumes/gen\_ai\_poc\_databrickscoe/sdlc\_wizard/customerdata` and load it to a Delta table named "customer".
2. Read order CSV data from `/Volumes/gen\_ai\_poc\_databrickscoe/sdlc\_wizard/orderdata` and load it to a Delta table named "order".
3. Add a new column "TotalAmount" to the "order" table, which is the product of "PricePerUnit" and "Qty" columns.

FR-DLT-002: Clean and Transform Customer and Order Data

Title: Clean and Transform Customer and Order Data

Description: The system must remove null and duplicate records from customer and order tables.

Preconditions:

- Customer and order Delta tables are loaded with data

Main Flow / Functional Steps:

1. Remove null records from both "customer" and "order" tables.
2. Remove duplicate records from both "customer" and "order" tables.

FR-DLT-003: Create ordersummary Table and Load Data

Title: Create ordersummary Table and Load Data

Description: The system must create an "ordersummary" table if it does not exist and load data into it by joining customer and order tables.

Preconditions:

- Customer and order Delta tables are cleaned and transformed
- "ordersummary" table does not exist or is empty

Main Flow / Functional Steps:

1. Create "ordersummary" table in catalog "gen_ai_poc_databrickscoe" and schema "sdlc_wizard" if it does not exist.
2. Join "customer" and "order" tables on "CustId" field.
3. Load the joined data into the "ordersummary" table as an SCD type 2 table.

FR-DLT-004: Update ordersummary Table on Customer Changes

Title: Update ordersummary Table on Customer Changes

Description: The system must update the "ordersummary" table whenever there is a change in the customer table.

Preconditions:

- "ordersummary" table is loaded with data
- Customer table is updated

Main Flow / Functional Steps:

1. Identify changes in the "customer" table.
2. Update the "ordersummary" table by making old records inactive and new records active.
3. Update "StartDate" and "EndDate" columns accordingly to maintain history.

FR-DLT-005: Create customeraggregatespend Table and Load Data

Title: Create customeraggregatespend Table and Load Data

Description: The system must create a "customeraggregatespend" table and load aggregated data into it.

Preconditions:

- "ordersummary" table is loaded with data

- "customeraggregatespend" table does not exist or is empty

Main Flow / Functional Steps:

1. Create "customeraggregatespend" table in catalog "gen_ai_poc_databrickscoe" and schema "sdlc_wizard" if it does not exist.
2. Aggregate "TotalAmount" column from "ordersummary" table by "Name" and "Date" columns.
3. Load the aggregated data into the "customeraggregatespend" table.

FR-DLT-006: Implement Delta Live Tables**Title: Implement Delta Live Tables**

Description: The system must implement Delta Live Tables to execute the above steps.

Preconditions:

- All previous steps are completed successfully

Main Flow / Functional Steps:

1. Implement Delta Live Tables to execute steps 1 to 5.

These functional requirements are based on the "Coding Requirements" section of the provided BRD.